# UWB Radar-Based Human Activity Recognition with CNNs: A Comparative Study Across Spatial Strategies and Window Lengths

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**Abstract.** Privacy concerns limit the adoption of video-based monitoring in smart homes. This study investigates Human Activity Recognition (HAR) in smart homes using Ultra-Wideband (UWB) radar sensing and deep learning, a non-intrusive and privacy-preserving modality. Experiments were conducted in a smart apartment with 19 participants performing 14 daily activities across five UWB-equipped rooms. Radar signals were preprocessed using three strategies: raw data, Chebyshev filtering, and moving-average smoothing, and segmented into 5 s, 10 s, and 15 s sliding windows with 90% overlap. We compare three classification strategies: apartment-level (C1), position-level (C2), and room-level (C3). Convolutional Neural Networks trained with Adadelta were evaluated using Top-$k$ accuracy, macro-F1, and Cohen's $\kappa$, with Top-1 results reported here for conciseness. Apartment-level models achieved moderate Top-1 performance (0.799 accuracy, $\kappa = 0.783$), with comparable trends observed for position-level models. In contrast, room-level models substantially outperformed them, reaching up to 1.000 accuracy ($\kappa = 1.000$) in the bathroom using a 5 s window with raw data, 0.998 in the dining room and 0.915 in the kitchen using a 15 s window with moving-average preprocessing, and 0.742 in the bedroom using a 15 s window with raw data. Performance varied across rooms: the living room peaked at 0.697 Top-1 accuracy with a 10 s window and moving-average preprocessing, reflecting increased activity diversity and spatial complexity. Longer windows (15 s) often improved recognition in several rooms, while shorter windows (5–10 s) were competitive or preferred depending on the room and preprocessing. These findings highlight the critical role of spatial context in radar-based HAR and demonstrate the effectiveness of UWB sensing combined with deep learning for privacy-preserving activity monitoring in smart-home and healthcare environments.

**Keywords:** Human Activity Recognition · Ultra-Wideband Radar · Deep Learning · Smart Homes · Temporal Windowing · Spatial Granularity.

## 1 INTRODUCTION

The rapid aging of the global population is intensifying pressure on healthcare systems, as longer life expectancy brings a higher prevalence of age-related con-

ditions such as Alzheimer's disease and other cognitive impairments [25]. At the same time, falls remain one of the leading causes of injury and loss of independence among older adults [26]. Together, these factors highlight the urgent need for proactive and continuous in-home monitoring solutions [26]. Sustaining Activities of Daily Living (ADLs) is tightly linked to quality of life and functional autonomy, making timely recognition of activity patterns and deviations a practical avenue for supporting independent living [4]. Beyond these clinical motivations, practical deployments also impose strict constraints on privacy. Elder-care monitoring systems must preserve residents' anonymity; compared to video-based methods, UWB radar avoids capturing identifiable imagery while still providing fine-grained spatiotemporal motion signatures [5, 10, 3]. These privacy considerations drive our systematic investigation of spatial granularity and temporal windowing as core design parameters for secure, real-world HAR systems. While this field has been explored across a spectrum of sensing modalities spanning ambient intelligence and the Internet of Things (IoT) [7], UWB remains distinct for its ability to balance fine-grained sensing with anonymity. Wearable inertial measurement units (IMUs) can capture fine-grained motion and have powered many successful systems, but they require user compliance and long-term acceptability that may be difficult to ensure in daily life [2]. Alternatives such as Wi-Fi sensing and ambient sensors provide environmental context with low intrusiveness, yet often lack the spatial and temporal resolution needed for fine-grained ADL discrimination [27]. Furthermore, generalization in HAR remains sensitive to dataset size, class balance, and protocol design, motivating rigorous evaluation in realistic, multi-participant settings [1, 8, 12]. To address this, cross-validation methods such as leave-one-subject-out (LOSO) have been proposed to ensure models are generalizable across individuals with different activity patterns [12]. LOSO cross-validation is particularly important in HAR research as it evaluates model performance on unseen participants, mimicking real-world deployment scenarios where the system must recognize activities for individuals not included in the training data. UWB radar has emerged as a compelling alternative for non-contact, privacy-preserving HAR in smart homes. By capturing range–time or range–Doppler signatures rather than identifiable imagery, UWB enables reliable operation under varied lighting and occlusion conditions while preserving user privacy [13, 19, 29]. Foundational studies have shown that radar sensing combined with deep learning can effectively recognize human motion in indoor environments [13, 18], while empirical works further demonstrate robust performance of deep networks trained on UWB signals in real smart apartment settings [19, 18]. Broader surveys and recent overviews highlight UWB's scalability and practicality within the ambient intelligence and healthcare landscape [29, 9]. This work addresses key deployment challenges related to spatial and temporal context. Spatially, activities are room-dependent, sensors have limited fields of view, and apartment-level models may be confounded by inter-room variability. Temporally, longer windows often enhance classification accuracy, while shorter windows improve responsiveness but may reduce separability. To study these trade-offs, we conducted a systematic experiment in a fully instrumented

smart apartment, where 19 participants performed 14 ADLs across five UWB-equipped rooms. Data were segmented into 5 s, 10 s, and 15 s sliding windows (90% overlap), and three spatial strategies were compared: (C1) apartment-level, (C2) position-level, and (C3) room-level. Convolutional Neural Networks (CNNs) were implemented in TensorFlow and trained using the Adadelta optimizer [30] with standard hyperparameters. Model performance was assessed across multiple runs using Top-$k$ accuracy and macro-F1 score [23], providing a robust evaluation of classification consistency and generalization. Within a unified protocol incorporating raw, Chebyshev-filtered, and moving-average preprocessing, we provide a comparative analysis of spatial granularity and examine the impact of temporal window size on classification accuracy. Empirically, room-level models consistently outperform global strategies, while the best window length and preprocessing are room-dependent: longer windows help in some rooms, whereas shorter or moderate windows are preferable in others. Overall, the findings reinforce UWB radar combined with deep learning as a robust, privacy-preserving foundation for HAR in smart homes [13, 19, 18, 29, 9].

## 2 Related Work

HAR using UWB radar has attracted increasing attention in recent years, particularly for smart-home applications where contactless sensing offers key advantages over wearables. Unlike inertial measurement units (IMUs), UWB radars capture fine-grained spatiotemporal motion without requiring body attachment, making them well suited for healthcare monitoring, ambient assisted living, and security applications [13, 19, 18]. A variety of machine learning and deep learning approaches have been proposed to improve the accuracy and robustness of radar-based HAR while addressing challenges such as multipath effects, noise, and high-dimensional signal representations [16].

Several studies have focused on reducing computational complexity to enable real-time deployment. Data preprocessing strategies such as selective bin filtering and denoising with convolutional autoencoders have proven effective for enhancing signal quality while maintaining efficiency. For instance, filtering irrelevant range bins can improve classification with lightweight models [11]; however, this work is limited to a small set of simple locomotion activities such as walking, sitting, and standing, which does not reflect the wider spectrum of ADLs encountered in real smart-home environments. Similarly, convolutional denoising autoencoders mitigate radar clutter and refine spatial structure, leading to more robust recognition [15]. However, that study focused on low-level motion patterns rather than diverse and functionally meaningful daily activities. These preprocessing approaches highlight optimization potential at the data level, particularly in constrained environments, but their restricted activity sets underscore the need for studying more realistic and heterogeneous ADL scenarios such as those targeted in our work.

Deep learning methods, especially CNNs, are dominant for UWB HAR [21]. CNNs are applied to time-range or Doppler representations to extract spatial fea-

tures, while hybrid architectures combine CNNs with recurrent layers to capture temporal dependencies. For example, a combined EfficientNet-LSTM pipeline demonstrated strong performance across multiple activities [16], underscoring the value of integrating temporal modeling with spatial feature extraction. However, their evaluation is conducted on a relatively small collection of dynamic gestures, typically between five and seven activities, which limits generalization to broader ADL contexts. Similarly, hybrid CNN-LSTM pipelines and ensemble strategies have shown robust recognition of ADLs in smart apartment settings [19, 18]. These studies primarily address common ADLs such as preparing food, washing dishes, eating, or watching TV. While relevant, these activity sets remain constrained and do not cover a wider range of behaviors such as hygiene-related actions, transitions between rooms, or fine-grained functional activities. Traditional machine learning remains relevant as well: Random Forest classifiers with engineered radar features can be competitive and interpretable in resource-limited scenarios [11, 6].

Beyond algorithmic advances, several works explore realistic deployments. Maitre *et al.* [18] investigated ADL recognition in a smart home setting with UWB and deep learning, demonstrating feasibility for daily monitoring. Brishtel *et al.* [5] extended UWB radar to in-cabin driving activity recognition and emphasized generalization to unseen individuals; their activity set is domain-specific and restricted to in-vehicle actions such as mirror checking, reaching, and glancing, which differs fundamentally from domestic ADLs encountered in smart-home scenarios. Such results reinforce that robust validation protocols like LOSO cross-validation are critical to avoid optimistic estimates [12]. Building on this, we also adopt LOSO for C1/C2 evaluation (with a participant-based 70%/15%/15% split for C3) to support cross-subject generalization to unseen participants, aligning our evaluation with best practices identified in prior HAR studies. In parallel, end-to-end learning on UWB channel impulse response snapshots has been shown competitive without handcrafted preprocessing [7].

Multi-sensor fusion is also explored to address single-sensor limitations: combining UWB with Wi-Fi/IMU sensing improves coverage and robustness while preserving low intrusiveness [27]. Together, these studies illustrate a progression from classical ML with handcrafted features [11] to deep neural architectures that extract spatiotemporal patterns directly from radar signals [21, 19]. Persistent challenges include ensuring cross-subject generalization [12], handling noisy real-world environments [15], and using evaluation metrics (e.g., top-$k$) that better capture robustness in multi-class scenarios [20].

Despite these contributions, most existing UWB HAR studies remain constrained by narrow activity sets or application-specific scenarios. Many works focus on simplified actions (e.g., walking, sitting, reach gestures) or domain-specific tasks (e.g., in-cabin monitoring), which do not reflect the diversity of complex ADLs performed across different rooms in a real apartment. In addition, prior studies typically evaluate a single spatial configuration or a single window length, making it difficult to understand how context resolution and temporal granularity jointly affect performance in realistic deployments.

Unlike prior studies that typically explore a single pipeline or focus narrowly on preprocessing enhancements [11, 15, 19, 18], our work provides a unified protocol that systematically varies both spatial granularity (apartment-, position-, and room-level) and temporal windowing (5, 10, 15 s) in the same smart apartment. This comparative design quantifies how context resolution and window duration jointly affect HAR performance, offering deployment guidelines beyond prior CNN–LSTM approaches.

Recent trends also push toward obstructed and distributed sensing. Through-the-wall UWB HAR [28] and distributed radar networks with time-frequency design choices [7] point to realistic and privacy-preserving monitoring. Similar to prior gesture-level approaches, their activity sets focus on short and isolated movements rather than a diverse collection of daily living activities. Meanwhile, architectures are expanding beyond pure CNNs, with transformer-based components explored to enhance generalization in distributed radar scenarios [17]. Overall, UWB HAR is evolving from controlled single-radar experiments toward realistic multi-radar and obstructed environments, with growing attention to spectral representation choices, sensor fusion, and modern deep learning backbones.

## 3 Methodology

### 3.1 Experimental Setup: Smart Apartment Environment

Data were collected in a real domestic environment instrumented across five rooms (bathroom, bedroom, dining room, kitchen, and living room) in the LIARA smart apartment. Nineteen participants performed 14 ADLs under a scripted but naturalistic protocol. The activities included *Brushing Teeth, Doing Housework, Drinking, Eating, Getting Dressed/Undressed, Going to the Toilet, Putting Away Clean Laundry, Putting Away Dishes, Reading a Book, Resting, Sleeping, Taking a Shower, Using a Computer/Smartphone, and Washing Dishes.*

UWB radars were installed to cover each room; signals were acquired with a frame rate of 50 Hz. Each frame comprises 368 range samples; following the established LIARA acquisition methodology, we retain the first 184 bins per frame for modeling. Activities were annotated with room labels, enabling apartment, sensor, and room-level analyses.

### 3.2 Dataset Preparation

Frames are associated with individual radars using serial numbers, grouped into per-sensor matrices (time $\times$ 184), and cropped to remove the first and last second of each recording. We prepare three preprocessing pipelines (described below) and serialize each windowed instance with its activity label for C1 and C2, and with both the activity label and the corresponding room label for C3. In addition to the participant-based train, validation, and test splits (70%, 15%, 15%) used for room-level (C3) evaluation, we apply a leave-one-subject-out (LOSO)

protocol for apartment-level (C1) and position-based (C2) evaluation to assess generalization to completely unseen individuals [12].

To facilitate deep learning, the continuous UWB recordings were segmented into discrete instances using a sliding window approach with 90% overlap [3]. This process resulted in a dense dataset; for example, the 15 s window configuration yielded a total of 16,612 instances across the 14 activities. Table 1 provides the specific distribution of these instances, illustrating the class balance within the dataset. As observed, the dataset is relatively well-balanced, which ensures that the high classification accuracies reported are not the result of a bias toward a dominant majority class [8].

**Table 1.** Distribution of learning instances (15 s windows, 90% overlap)

| **Activity Class** | **N** | **Activity Class** | **N** |
|---|---|---|---|
| Brushing Teeth | 1,178 | Put. Away Dishes | 1,178 |
| Doing Housework | 818 | Reading Book | 1,353 |
| Drinking | 1,178 | Resting | 1,178 |
| Eating | 1,353 | Sleeping | 1,351 |
| Get. Dressed | 799 | Taking Shower | 1,056 |
| Going to Toilet | 1,118 | Using Smartphone | 1,351 |
| Put. Away Laundry | 1,352 | Washing Dishes | 1,349 |
| Total Instances | | | 16,612 |

### 3.3 Preprocessing Pipelines

We compare three preprocessing variants:

- *Raw*: cropping only, no filtering.
- *Chebyshev band-pass*: a Type-I filter (order 2, passband 1–5 Hz, 0.025 dB ripple) implemented in second-order sections to attenuate low-frequency drift and high-frequency noise [3].
- *Moving-average removal*: subtraction of a rolling mean over the previous 50 frames to suppress slowly varying backgrounds.

Unless otherwise stated, moving-average removal is used as the default preprocessing method in all experiments.

### 3.4 Segmentation and Labeling

Signals are segmented into sliding windows of 5, 10, or 15 s with 90% overlap. A window inherits the activity label if the annotated activity occupies the majority of its duration. Window tensors have shape $T \times 184 \times R$, where $T$ is the number of frames in the window (e.g., $T = 750$ for 15 s at 50 Hz) and $R$ is the number of radars included by the selected spatial strategy.

### 3.5 Classification Strategies

We study three spatial granularities on the same recordings. **C1 (apartment-level)** concatenates all available sensors to train a single model representing the entire apartment. **C2 (position-based)** performs classification according to the participant's location within the environment: for each window, data from the one UWB radar closest to the participant is selected, and a model is trained using only those signals. In this study, "closest" is determined from annotated participant location (oracle proximity); C2 is therefore an upper-bound localized-sensing setup and would require an online localization or radar-selection stage for fully blind deployment. This strategy reflects localized sensing around the person's position and enables analysis of how proximity and radar geometry influence recognition accuracy. **C3 (room-level)** trains a separate model for each room, using only the samples associated with activities that took place in that room, thus leveraging room-dependent contextual information.

### 3.6 Model Architecture

Our primary network for 15 s windows learns directly from time–range inputs and follows a compact convolutional design. Beginning with batch normalization [14], the model applies a convolutional layer with 32 filters using a $7 \times 7$ kernel at stride 2, followed by a rectified linear activation and a second normalization step. Two additional convolutional layers with 64 filters ($5 \times 5$ kernels, stride 2) extract mid-level spatial features, again followed by batch normalization. Deeper layers increase representational capacity using 128 filters with $3 \times 3$ kernels (stride 2), interleaved with batch normalization, culminating in a final $3 \times 3$ convolution and normalization block. The resulting feature maps are flattened and passed through two fully connected layers with 200 and 100 units (ReLU activations), followed by a fourteen-class softmax output.

For shorter temporal contexts, compact network variants are employed that preserve the same architectural principles while explicitly reducing model complexity. These models retain a convolutional feature extractor with batch normalization but use fewer convolutional layers, reduced channel widths, and earlier feature flattening, resulting in a substantially lower parameter count. Regularization is introduced through dropout applied in the fully connected classification head [24], with layer widths and dropout rates empirically tuned using validation performance to mitigate overfitting. In contrast, a deeper high-capacity architecture stacks repeated convolution, batch-normalization, and max-pooling blocks with progressively increasing channel dimensionality up to 512 filters, followed by a dropout-regularized classification head, providing a capacity-increased baseline for comparison.

### 3.7 Training and Optimization

All CNNs are implemented in TensorFlow/Keras and trained for 50 epochs with a batch size of 16 using the Adadelta optimizer (learning rate 0.1) [30] and categorical cross-entropy loss. Validation accuracy is monitored during training, and

the model achieving the highest validation performance is retained for evaluation. To account for stochastic variability inherent to neural network training, each experimental configuration defined by the preprocessing method, temporal window length, and spatial classification strategy is repeated across 10 independent runs. Final results are reported as the mean ± standard deviation.

### 3.8 Evaluation Protocol and Metrics

For testing, we re-instantiate the corresponding architecture, load the best-validation checkpoint, and compute Top-$k$ accuracy ($k \in \{1, 2, 3, 4, 5\}$) [20], macro-F1 and balanced accuracy. Macro-F1 and balanced accuracy are used to handle potential class imbalance, following standard recommendations in classification evaluation [23]. Window sizes of 5, 10, and 15 s follow prior studies on temporal segmentation trade-offs [3]. All evaluations keep subjects disjoint across train, validation, and test to support cross-subject generalization. For C3, results are reported per room and aggregated where appropriate. All runs export logs, learning curves, and model artifacts for reproducibility; consolidated metrics are stored in structured spreadsheets for statistical analysis. Reported scores use strategy-specific participant splits, with no subject overlap between train, validation, and test. For apartment-level (C1) and position-based (C2) models, tabulated results follow leave-one-subject-out (LOSO) evaluation over the 19 participants (190 subject×run cycles for C1). For room-level (C3) models, tabulated results follow a participant-based 70%/15%/15% split (13/3/3 subjects); each room model is trained and evaluated only on that room's activity label set. In all cases, each configuration is repeated for 10 independent training runs, and we report Top-1 mean ± standard deviation across these runs. For C3 only, we additionally compared paired Top-1 accuracies across the same ten independent runs using two-sided Wilcoxon signed-rank tests at $\alpha = 0.05$; these tests do not cover C1/C2.

## 4 Results

We evaluated our UWB-based CNN models under three spatial strategies: (C1) apartment-level, (C2) position-based, and (C3) room-level, using temporal windows of 5 s, 10 s, and 15 s. Models were assessed with Top-$k$ accuracy [20], macro-F1 and balanced accuracy. Window lengths were chosen to investigate the influence of temporal duration on recognition performance [3]. Tabulated C1/C2 scores use LOSO, while tabulated C3 scores use a participant-based 70%/15%/15% split (13/3/3 subjects); in all cases subjects remain disjoint across train, validation, and test [12].

### 4.1 Whole-apartment classification (C1)

The evaluation utilized a Leave-One-Subject-Out (LOSO) scheme across 190 cycles. For apartment-level (C1) classification, raw data with 5-second windows

yielded moderate results ($0.619 \pm 0.026$ accuracy), struggling with low-motion activities. However, moving average preprocessing significantly enhanced performance, peaking at 10 seconds with $0.799 \pm 0.025$ Top-1 accuracy and $0.807 \pm 0.025$ balanced accuracy. In contrast, raw data degraded at 15 seconds ($0.617 \pm 0.051$ accuracy), whereas filtered methods remained robust. This confirms that signal smoothing is vital for stability, with the 10-second window offering the best trade-off between resolution and predictive power, as shown in Table 2.

**Table 2.** Classification 1 performance summary (Top-1 mean ± SD).

| Time Window | Method | Accuracy | Bal. Acc. | F1-score | Kappa |
|---|---|---|---|---|---|
| 5 sec | Raw Data | $0.619 \pm 0.026$ | $0.609 \pm 0.035$ | $0.547 \pm 0.037$ | $0.589 \pm 0.029$ |
| | Chebyshev | $0.758 \pm 0.015$ | $0.766 \pm 0.014$ | $0.752 \pm 0.013$ | $0.739 \pm 0.016$ |
| | Mov. Avg. | $0.750 \pm 0.026$ | $0.757 \pm 0.025$ | $0.749 \pm 0.023$ | $0.730 \pm 0.028$ |
| 10 sec | Raw Data | $0.649 \pm 0.024$ | $0.638 \pm 0.030$ | $0.595 \pm 0.034$ | $0.621 \pm 0.026$ |
| | Chebyshev | $0.755 \pm 0.011$ | $0.762 \pm 0.014$ | $0.752 \pm 0.012$ | $0.735 \pm 0.012$ |
| | Mov. Avg. | $0.799 \pm 0.025$ | $0.807 \pm 0.025$ | $0.798 \pm 0.023$ | $0.783 \pm 0.027$ |
| 15 sec | Raw Data | $0.617 \pm 0.051$ | $0.596 \pm 0.054$ | $0.568 \pm 0.056$ | $0.587 \pm 0.055$ |
| | Chebyshev | $0.768 \pm 0.017$ | $0.767 \pm 0.020$ | $0.760 \pm 0.016$ | $0.750 \pm 0.019$ |
| | Mov. Avg. | $0.773 \pm 0.032$ | $0.785 \pm 0.030$ | $0.765 \pm 0.031$ | $0.756 \pm 0.034$ |

Furthermore, while Chebyshev filtering also provided strong improvements over raw data ($0.768 \pm 0.017$ accuracy at 15 s), the moving average strategy proved marginally superior and more consistent. This indicates that suppressing static background clutter via simple subtraction is highly effective for UWB-based activity recognition, minimizing the impact of environmental noise.

### 4.2 Position-based (Classification 2) performance

As presented in Table 3, classification performance varied notably across configurations, with clear benefits observed from signal filtering. The best overall results were obtained utilizing Chebyshev and moving-average preprocessing at the 15 s window, where both methods reached peak accuracies of $0.807 \pm 0.017$ and $0.807 \pm 0.032$, respectively. Beyond standard accuracy, the macro-F1 scores highlight critical differences in model reliability and class balance. Short 5 s windows provided reasonable baselines, with Raw Data yielding an F1-score of $0.673 \pm 0.050$, whereas the Moving Average method improved this significantly to $0.772 \pm 0.019$. This trend of stability in filtered data persists as the window size increases.

At 10 s, the Moving Average approach achieved a strong F1-score of $0.792 \pm 0.027$, indicating consistent recognition across diverse activities. Conversely, the limitations of unprocessed signals became evident at the 15 s duration; while filtered methods maintained high F1-scores (approximately 0.798), Raw Data degraded significantly to an F1-score of $0.587 \pm 0.056$. These findings confirm

**Table 3.** Average position-based (classification 2) performance across all configurations (Top-1 mean ± SD).

| Time Window | Method | Accuracy | Bal. Acc. | F1-score | Kappa |
|---|---|---|---|---|---|
| 5 sec | Raw Data | $0.704 \pm 0.047$ | $0.715 \pm 0.044$ | $0.673 \pm 0.050$ | $0.681 \pm 0.051$ |
| | Chebyshev | $0.768 \pm 0.015$ | $0.774 \pm 0.014$ | $0.762 \pm 0.011$ | $0.750 \pm 0.016$ |
| | Mov. Avg. | $0.776 \pm 0.021$ | $0.778 \pm 0.020$ | $0.772 \pm 0.019$ | $0.759 \pm 0.022$ |
| 10 sec | Raw Data | $0.721 \pm 0.036$ | $0.724 \pm 0.036$ | $0.691 \pm 0.034$ | $0.699 \pm 0.038$ |
| | Chebyshev | $0.784 \pm 0.030$ | $0.797 \pm 0.032$ | $0.778 \pm 0.032$ | $0.767 \pm 0.033$ |
| | Mov. Avg. | $0.797 \pm 0.030$ | $0.797 \pm 0.031$ | $0.792 \pm 0.027$ | $0.781 \pm 0.033$ |
| 15 sec | Raw Data | $0.639 \pm 0.051$ | $0.619 \pm 0.054$ | $0.587 \pm 0.056$ | $0.610 \pm 0.055$ |
| | Chebyshev | $0.807 \pm 0.017$ | $0.807 \pm 0.020$ | $0.797 \pm 0.016$ | $0.791 \pm 0.019$ |
| | Mov. Avg. | $0.807 \pm 0.032$ | $0.811 \pm 0.030$ | $0.798 \pm 0.031$ | $0.792 \pm 0.034$ |

that leveraging the closest radar position yields strong, noise-robust recognition, but only when temporal smoothing is applied to mitigate the drift and variability inherent in longer raw signal windows [3].

### 4.3 Per-room classification (C3)

Overall, the room-level strategy (C3) substantially outperforms both the apartment-level (C1) and position-level (C2) approaches. This performance gap is primarily driven by the exploitation of room semantics, which provide strong spatial priors that effectively constrain the classification problem [5, 10]. Unlike global models that must discriminate between all possible activities simultaneously, the C3 strategy reduces inter-room ambiguity by isolating the activity set relevant to each specific environment which is a pattern consistent with established findings in ambient intelligence.

In practice, restricting inference to a single room drastically narrows the candidate classes (e.g., excluding 'cooking' when analyzing Bedroom data), thereby simplifying the decision boundary for the classifier. Furthermore, this approach ensures that the sensing geometry is consistently aligned with location-specific behaviors, maximizing the capture of relevant motion features. These advantages are quantitatively reflected in our confusion matrices, which exhibit significantly fewer cross-room confusions compared to the global strategies. When comparing C3 to C1, note that apartment-level models solve a 14-class problem, whereas each room-level model discriminates among a smaller, room-specific label set (e.g., $K$=3 in the bathroom, $K$=8 in the bedroom, and $K$=5 in the living room). Higher C3 accuracy therefore reflects both spatial context and a reduced decision space; Cohen's $\kappa$ mitigates chance agreement but does not fully equalize task difficulty across different $K$. Wilcoxon signed-rank tests on the ten C3 runs support the main per-room contrasts ($p$=0.002 unless noted). In the living room, 10 s moving average significantly outperformed 10 s Chebyshev and 15 s moving average, but not 15 s Chebyshev ($p$=0.322). In the bedroom, 15 s raw and moving average both significantly exceeded 15 s Chebyshev, while raw

vs. moving average was not significant ($p$=0.065). In the bathroom, 5 s raw was significantly higher than 5 s and 15 s moving average. In the dining room, 15 s moving average significantly exceeded 10 s moving average and 15 s raw. In the kitchen, 15 s moving average beat 15 s Chebyshev, but 10 s moving average was significantly higher still.

**Bathroom classification performance** The Bathroom-specific classification achieved near-perfect results, suggesting distinct and highly recognizable activity patterns within this environment.

As detailed in Table 4, the shortest window (5 s) yielded an exceptional $1.000 \pm 0.000$ for both accuracy and F1-score with Raw Data. Filtered methods at this duration were slightly less effective; specifically, the Moving Average method achieved an accuracy of $0.920 \pm 0.004$ and an F1-score of $0.909 \pm 0.004$, indicating a minor loss in precision compared to the raw signal. Performance trends shifted slightly as the temporal window expanded. At 10 s, Raw Data maintained its dominance with $0.999 \pm 0.001$ across all metrics.

However, the benefits of smoothing became apparent at the 15 s interval. While Raw Data experienced a slight increase in variability (F1-score SD rising to 0.034), the Moving Average method reached near-perfection with an accuracy of $0.994 \pm 0.003$ and an F1-score of $0.996 \pm 0.002$. Conversely, Chebyshev filtering showed signs of degradation at this longer interval, dropping to an F1-score of $0.953 \pm 0.003$.

These results suggest that while the bathroom environment allows for reliable recognition even with raw signals, moving-average smoothing offers superior stability for extended temporal windows.

**Table 4.** Bathroom classification performance across time windows (Top-1 mean ± SD).

| Time Window | Method | Accuracy | Bal. Acc. | F1-score | Kappa |
|---|---|---|---|---|---|
| 5 sec | Raw Data | $1.000 \pm 0.000$ | $1.000 \pm 0.000$ | $1.000 \pm 0.000$ | $1.000 \pm 0.000$ |
| | Chebyshev | $0.977 \pm 0.019$ | $0.983 \pm 0.014$ | $0.979 \pm 0.017$ | $0.965 \pm 0.029$ |
| | Mov. Avg. | $0.920 \pm 0.004$ | $0.935 \pm 0.003$ | $0.909 \pm 0.004$ | $0.879 \pm 0.006$ |
| 10 sec | Raw Data | $0.999 \pm 0.001$ | $0.999 \pm 0.001$ | $0.999 \pm 0.001$ | $0.999 \pm 0.001$ |
| | Chebyshev | $0.992 \pm 0.002$ | $0.995 \pm 0.001$ | $0.994 \pm 0.002$ | $0.988 \pm 0.003$ |
| | Mov. Avg. | $0.975 \pm 0.010$ | $0.967 \pm 0.016$ | $0.965 \pm 0.014$ | $0.959 \pm 0.016$ |
| 15 sec | Raw Data | $0.989 \pm 0.026$ | $0.988 \pm 0.031$ | $0.987 \pm 0.034$ | $0.984 \pm 0.041$ |
| | Chebyshev | $0.947 \pm 0.004$ | $0.952 \pm 0.003$ | $0.953 \pm 0.003$ | $0.918 \pm 0.006$ |
| | Mov. Avg. | $0.994 \pm 0.003$ | $0.996 \pm 0.002$ | $0.996 \pm 0.002$ | $0.992 \pm 0.004$ |

**Bedroom classification performance** Bedroom-specific classification performance varied notably across pre-processing methods and temporal windows. As

detailed in Table 5, the 5 s window yielded the lowest results, with accuracies ranging from $0.461 \pm 0.024$ for raw data to $0.600 \pm 0.027$ using a Moving Average.

At 10 s, Moving Average reached a strong $0.734 \pm 0.012$ accuracy and Chebyshev improved to $0.692 \pm 0.014$, but the overall best Top-1 accuracy was obtained with Raw Data at 15 s ($0.742 \pm 0.044$). At the same 15 s window, Chebyshev collapsed to $0.131 \pm 0.081$ ($p$=0.002 vs. 15 s raw and vs. 15 s moving average; Wilcoxon), while Moving Average remained competitive ($0.700 \pm 0.021$); raw vs. moving average at 15 s was not significant ($p$=0.065). Overall, bedroom activities, which often involve subtle movements, can benefit from denoising, but filter choice is configuration-dependent: Moving Average is more stable across windows than Chebyshev, whereas the single best accuracy came from longer-window Raw Data.

This 15 s Chebyshev collapse ($0.131 \pm 0.081$ accuracy; $\kappa = 0.003 \pm 0.009$) is near chance for the bedroom's eight-class problem and is configuration-specific: the same 15 s Chebyshev setting remains competitive in the living room ($0.685 \pm 0.034$), while bedroom 15 s Raw and Moving Average stay strong ($0.742 \pm 0.044$ and $0.700 \pm 0.021$). A plausible explanation is that the fixed 1–5 Hz Type-I bandpass removes motion cues that matter for subtle bedroom activities once windows become long, whereas Moving Average background subtraction preserves usable structure. We therefore treat Chebyshev as brittle for long-window bedroom deployment and recommend validating filter choice per room rather than assuming a single preprocessing default.

**Table 5.** Bedroom classification performance across time windows (Top-1 mean ± SD).

| Time Window | Method | Accuracy | Bal. Acc. | F1-score | Kappa |
|---|---|---|---|---|---|
| 5 sec | Raw Data | $0.461 \pm 0.024$ | $0.347 \pm 0.041$ | $0.327 \pm 0.041$ | $0.348 \pm 0.030$ |
| | Chebyshev | $0.589 \pm 0.079$ | $0.422 \pm 0.072$ | $0.405 \pm 0.072$ | $0.484 \pm 0.097$ |
| | Mov. Avg. | $0.600 \pm 0.027$ | $0.508 \pm 0.046$ | $0.491 \pm 0.036$ | $0.526 \pm 0.033$ |
| 10 sec | Raw Data | $0.474 \pm 0.025$ | $0.356 \pm 0.048$ | $0.337 \pm 0.047$ | $0.363 \pm 0.031$ |
| | Chebyshev | $0.692 \pm 0.014$ | $0.588 \pm 0.024$ | $0.558 \pm 0.024$ | $0.632 \pm 0.017$ |
| | Mov. Avg. | $0.734 \pm 0.012$ | $0.677 \pm 0.014$ | $0.663 \pm 0.021$ | $0.687 \pm 0.014$ |
| 15 sec | Raw Data | $0.742 \pm 0.044$ | $0.593 \pm 0.030$ | $0.572 \pm 0.040$ | $0.686 \pm 0.050$ |
| | Chebyshev | $0.131 \pm 0.081$ | $0.128 \pm 0.007$ | $0.035 \pm 0.021$ | $0.003 \pm 0.009$ |
| | Mov. Avg. | $0.700 \pm 0.021$ | $0.557 \pm 0.026$ | $0.539 \pm 0.026$ | $0.625 \pm 0.027$ |

**Dining room classification performance** Dining room classification achieved strong to near-perfect performance depending on preprocessing and window length. As shown in Table 6, the 5 s window already yields high accuracy with Raw Data ($0.917 \pm 0.001$), though filtered methods start more modestly, with Chebyshev at $0.701 \pm 0.052$ and Moving Average at $0.644 \pm 0.036$. As the temporal window extends, the performance dynamics shift. At 10 s, the Chebyshev filter shows a significant leap to $0.915 \pm 0.009$ accuracy, while Raw Data accuracy

temporarily dips to $0.739 \pm 0.060$. The most definitive results occur at the 15 s window, where the model achieves near-perfect classification using Moving Average preprocessing, reaching $0.998 \pm 0.001$ accuracy and $0.998 \pm 0.001$ Balanced Accuracy.

These trends suggest that repetitive dining room activities are best captured over extended periods. While raw signals suffice for short windows, Moving Average smoothing virtually eliminates classification errors at longer windows. Two metric patterns deserve comment. First, Moving Average balanced accuracy jumps from $0.621 \pm 0.012$ at 10 s to $0.998 \pm 0.001$ at 15 s, suggesting longer windows better capture repetitive dining motions and reduce transition ambiguity. Second, Accuracy and Balanced Accuracy can diverge sharply at room level (e.g., 10 s Chebyshev: 0.915 vs. 0.725; 15 s Raw: 0.870 vs. 0.949), so class-balance arguments based on apartment-wide counts do not automatically protect per-room Accuracy; Balanced Accuracy and macro-F1 remain necessary.

**Table 6.** Dining Room classification performance across time windows (Top-1 mean ± SD).

| Time Window | Method | Accuracy | Bal. Acc. | F1-score | Kappa |
|---|---|---|---|---|---|
| 5 sec | Raw Data | $0.917 \pm 0.001$ | $0.916 \pm 0.001$ | $0.864 \pm 0.001$ | $0.861 \pm 0.002$ |
| | Chebyshev | $0.701 \pm 0.052$ | $0.603 \pm 0.035$ | $0.578 \pm 0.047$ | $0.533 \pm 0.082$ |
| | Mov. Avg. | $0.644 \pm 0.036$ | $0.618 \pm 0.060$ | $0.575 \pm 0.037$ | $0.363 \pm 0.061$ |
| 10 sec | Raw Data | $0.739 \pm 0.060$ | $0.779 \pm 0.051$ | $0.746 \pm 0.076$ | $0.454 \pm 0.138$ |
| | Chebyshev | $0.915 \pm 0.009$ | $0.725 \pm 0.016$ | $0.709 \pm 0.013$ | $0.846 \pm 0.017$ |
| | Mov. Avg. | $0.804 \pm 0.012$ | $0.621 \pm 0.012$ | $0.624 \pm 0.018$ | $0.679 \pm 0.021$ |
| 15 sec | Raw Data | $0.870 \pm 0.002$ | $0.949 \pm 0.001$ | $0.879 \pm 0.003$ | $0.788 \pm 0.002$ |
| | Chebyshev | $0.811 \pm 0.024$ | $0.661 \pm 0.022$ | $0.660 \pm 0.028$ | $0.674 \pm 0.042$ |
| | Mov. Avg. | $0.998 \pm 0.001$ | $0.998 \pm 0.001$ | $0.997 \pm 0.002$ | $0.997 \pm 0.002$ |

**Kitchen classification performance** Kitchen classification performance remained high across most preprocessing methods, with filtered data consistently outperforming raw inputs. As shown in Table 7, the 5 s window already provides strong results, particularly with the moving average smoothing which achieved an accuracy of $0.942 \pm 0.010$. During this same interval, the Chebyshev filter also performed well at $0.921 \pm 0.002$, while raw data lagged significantly at $0.644 \pm 0.056$.

As the temporal window increased to 10 s, performance remained relatively stable for filtered methods. The moving average maintained an accuracy of $0.940 \pm 0.006$, while the Chebyshev filter saw a slight decrease to $0.900 \pm 0.006$. Interestingly, the 15 s configuration did not yield the "near perfect" results suggested in previous environments; instead, accuracy for the moving average filter dropped slightly to $0.915 \pm 0.005$, and the Chebyshev filter fell to $0.856 \pm 0.014$.

Raw data showed a minor recovery at this longer window, reaching $0.679 \pm 0.104$ accuracy.

These results suggest that kitchen activities are best captured at shorter to moderate temporal scales. The moving average filter proved to be the most robust method, effectively managing the sensor noise. However, the slight decline in performance at 15 s indicates that longer windows may introduce excessive signal overlap or unrelated transitions that slightly degrade classification precision.

**Table 7.** Kitchen classification performance across time windows (Top-1 mean ± SD).

| Time Window | Method | Accuracy | Bal. Acc. | F1-score | Kappa |
|---|---|---|---|---|---|
| 5 sec | Raw Data | $0.644 \pm 0.056$ | $0.550 \pm 0.087$ | $0.536 \pm 0.084$ | $0.535 \pm 0.072$ |
| | Chebyshev | $0.921 \pm 0.002$ | $0.816 \pm 0.003$ | $0.817 \pm 0.005$ | $0.894 \pm 0.003$ |
| | Mov. Avg. | $0.942 \pm 0.010$ | $0.875 \pm 0.020$ | $0.891 \pm 0.015$ | $0.910 \pm 0.015$ |
| 10 sec | Raw Data | $0.531 \pm 0.051$ | $0.489 \pm 0.076$ | $0.428 \pm 0.079$ | $0.401 \pm 0.063$ |
| | Chebyshev | $0.900 \pm 0.006$ | $0.778 \pm 0.031$ | $0.760 \pm 0.015$ | $0.868 \pm 0.008$ |
| | Mov. Avg. | $0.940 \pm 0.006$ | $0.887 \pm 0.023$ | $0.894 \pm 0.011$ | $0.906 \pm 0.010$ |
| 15 sec | Raw Data | $0.679 \pm 0.104$ | $0.614 \pm 0.108$ | $0.551 \pm 0.112$ | $0.574 \pm 0.128$ |
| | Chebyshev | $0.856 \pm 0.014$ | $0.831 \pm 0.010$ | $0.806 \pm 0.018$ | $0.813 \pm 0.018$ |
| | Mov. Avg. | $0.915 \pm 0.005$ | $0.842 \pm 0.006$ | $0.817 \pm 0.011$ | $0.885 \pm 0.007$ |

**Living room classification performance** Living room classification varied significantly by configuration (Table 8). At 5 s, Moving Average already outperformed Raw Data and Chebyshev ($0.670 \pm 0.063$) and reached its peak at 10 s with $0.697 \pm 0.023$ accuracy ($p$=0.002 vs. 10 s Chebyshev and vs. 15 s moving average; Wilcoxon), accompanied by the strongest balanced accuracy, F1-score, and Kappa among all living-room settings. Extending the window to 15 s did not yield a further gain for Moving Average, which dropped sharply to $0.329 \pm 0.108$, whereas Chebyshev remained competitive ($0.685 \pm 0.034$) and Raw Data stayed near its mid-range performance ($0.567 \pm 0.053$). Overall, living-room activities, often involving mixed postures, transitions, and interactions with furniture, benefited most from a moderate temporal context combined with moving-average smoothing, which mitigates transient noise while avoiding the degradation observed for Moving Average at the longest window.

**Table 8.** Living room classification performance across time windows (Top-1 mean ± SD).

| Time Window | Method | Accuracy | Bal. Acc. | F1-score | Kappa |
|---|---|---|---|---|---|
| 5 sec | Raw Data | 0.345 ± 0.034 | 0.367 ± 0.040 | 0.305 ± 0.041 | 0.198 ± 0.040 |
| | Chebyshev | 0.440 ± 0.026 | 0.391 ± 0.033 | 0.386 ± 0.030 | 0.279 ± 0.036 |
| | Mov. Avg. | 0.670 ± 0.063 | 0.589 ± 0.066 | 0.549 ± 0.063 | 0.555 ± 0.083 |
| 10 sec | Raw Data | 0.588 ± 0.012 | 0.408 ± 0.018 | 0.360 ± 0.033 | 0.387 ± 0.018 |
| | Chebyshev | 0.461 ± 0.017 | 0.526 ± 0.032 | 0.454 ± 0.022 | 0.304 ± 0.026 |
| | Mov. Avg. | 0.697 ± 0.023 | 0.688 ± 0.023 | 0.683 ± 0.026 | 0.606 ± 0.031 |
| 15 sec | Raw Data | 0.567 ± 0.053 | 0.408 ± 0.062 | 0.407 ± 0.087 | 0.401 ± 0.086 |
| | Chebyshev | 0.685 ± 0.034 | 0.576 ± 0.040 | 0.565 ± 0.040 | 0.565 ± 0.050 |
| | Mov. Avg. | 0.329 ± 0.108 | 0.375 ± 0.033 | 0.350 ± 0.043 | 0.108 ± 0.064 |

### 4.4 Confusion matrix insights

We use confusion matrices to examine error modes beyond aggregate metrics [23]. As shown in Fig. 1, the living-room model recognizes using a computer/smartphone perfectly (1.00), but struggles with other sedentary activities: doing housework is mostly predicted as resting (0.59) or smartphone use (0.33), and drinking is never correctly predicted (0.00), being absorbed by resting (0.74) and smartphone use (0.26). In contrast, Fig. 2 shows that in the bedroom, doing housework, putting away clean laundry, and sleeping are recognized perfectly (1.00); however, static postures remain difficult: using a computer/smartphone is almost entirely predicted as reading a book (0.97), and getting dressed/undressed is frequently confused with putting away laundry (0.51) is almost entirely predicted as reading a book (0.97), and getting dressed/undressed is frequently confused with putting away laundry (0.51). These patterns align with our quantitative findings and support room-level (C3) modeling.

## 5 Discussion

Our evaluation shows that spatial granularity and temporal window length jointly shape UWB radar-based HAR. Room-level models benefit from room semantics, but part of the C3 gain is a smaller label space; living-room performance remains hardest (best Acc 0.697 at 10 s Moving Average), while bathroom and dining can approach ceiling accuracy. Longer windows help in some rooms yet hurt in others (e.g., living-room 15 s Moving Average drops to 0.329; bedroom 15 s Chebyshev collapses to 0.131), so window and preprocessing should be tuned per room rather than fixed globally. For deployment, room-aware models suit privacy-preserving in-home monitoring, but transfer across apartments and sensor layouts remains open; LOSO (C1/C2) stresses unseen subjects, whereas C3's 70%/15%/15% split should be read as subject-disjoint holdout rather than fully unseen-subject LOSO. Cross-environment transfer remains challenging, motivating multimodal UWB–IMU fusion and adaptive windowing in future work [12, 3, 27, 22].

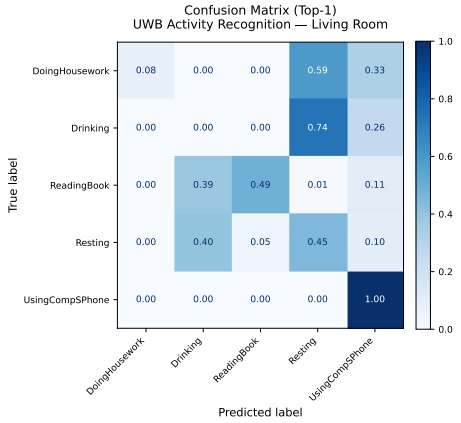



**Fig. 1.** Row-normalized Top-1 confusion matrix for living-room (C3) classification (10 s window, moving-average preprocessing, participant-based 70%/15%/15% split; representative run).

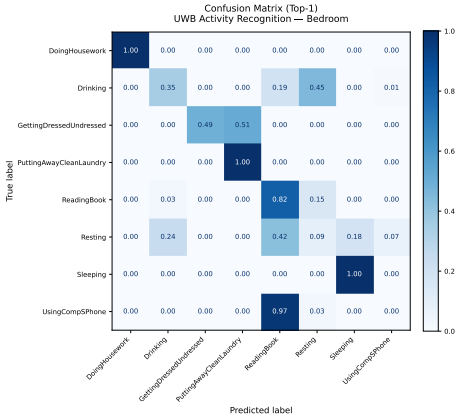



**Fig. 2.** Row-normalized Top-1 confusion matrix for bedroom (C3) classification (15 s window, raw preprocessing, participant-based 70%/15%/15% split; representative run).

A limitation of the C3-versus-C1 comparison is that room models operate on restricted label spaces. Separating spatial context from label-space reduction would require a controlled evaluation, e.g., scoring C1 with predictions restricted to each room's classes, or a two-stage pipeline (room selection from radar activation, then the room model) reported end-to-end including room-selection errors. We leave this controlled comparison to future work and interpret C3 gains accordingly.

# 6 CONCLUSIONS AND FUTURE WORKS

## 6.1 Conclusions

The presented experiments provide a comprehensive quantitative and qualitative assessment of spatial and temporal trade-offs in UWB-based activity recognition. This work presented a systematic study of UWB radar–based HAR in a real smart apartment, jointly varying spatial granularity and temporal windowing under a unified, reproducible protocol. Across nineteen participants and fourteen ADLs, room-level models (C3) consistently outperformed apartment-level (C1) and position-level (C2) approaches, confirming that spatial context, i.e., aligning models with the semantics of each room, substantially improves recognition. Per-room performance spanned a wide range (e.g., dining room high, living room comparatively lower), reflecting differences in activity mix, overlap, and field-of-view constraints. In the temporal dimension, the best window length and preprocessing pipeline were room-dependent: longer windows (15 s) improved recognition in several rooms (e.g., dining with moving average; bedroom with raw data), whereas shorter windows (5–10 s) were preferred in others (e.g., bathroom at 5 s raw; living room at 10 s moving average; kitchen at 5–10 s), suggesting that deployments should tune window length and filtering per room rather than adopting a single global setting. These findings held across three preprocessing pipelines (raw, Chebyshev, moving-average) and subject-disjoint splits (LOSO for C1/C2; 70%/15%/15% for C3) designed to prevent identity leakage, reinforcing the practicality of non-contact, privacy-preserving UWB sensing for continuous in-home monitoring.

## 6.2 Future Works

Future work will focus on the fusion of UWB radar and wearable IMU data using deep learning to jointly exploit spatial and motion cues for robust activity recognition. We will also investigate classical and hybrid baselines (Random Forest, SVM, and CNN–LSTM) with associated computational cost metrics, as well as cross-apartment transfer, adaptive windowing, lightweight models for real-time on-device inference, and controlled C3-versus-C1 comparisons that isolate label-space effects from spatial context. Taken together, the evidence indicates that pairing UWB sensing with modern deep learning, and extending it through multimodal fusion, offers a powerful foundation for practical, privacy-preserving HAR in smart environments.

# ETHICAL IMPACT STATEMENT

This work investigates Human Activity Recognition (HAR) using UWB radar, a sensing modality selected specifically for its privacy-preserving advantages over optical cameras. Unlike video-based systems, UWB sensors capture radio-frequency reflections rather than identifiable visual imagery, significantly mitigating privacy risks in sensitive domestic environments like bathrooms and bedrooms. Building on this secure foundation, data collection was conducted in the

LIARA smart apartment under an ethics-approved protocol with informed consent from all 19 participants. To further protect identity during analysis, all recordings were anonymized. Evaluation used LOSO for apartment-level (C1) and position-based (C2) models, and a participant-based 70%/15%/15% split for room-level (C3) models.

While the primary application of this research is to support independent living and elderly care through non-intrusive monitoring, we acknowledge that high-precision localization technologies carry potential risks if repurposed for unauthorized surveillance. To address these concerns, future real-world deployments must prioritize secure data encryption and edge-processing paradigms that prevent the transmission of raw signals. Additionally, although our current dataset includes 19 diverse participants, we recognize the critical importance of continuously expanding demographic representation in training data to prevent algorithmic bias across different age groups and mobility levels.

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